

Advanced (Habanero)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $x(x + 4) + 5(x + 4)$
- (B) $x(x + 5) + 4(x + 5)$
- (C) $x^2 + 9x + 20$
- (D) $x(x + 4) + 5(x + 4)$
- (E) $(x + 4)(x + 5)$

Q2. Factor: $3x^2 + 12x + 4x + 16$

- (A) $3x(x + 4) + 4(x + 4)$
- (B) $3x(x + 4) + 4(x - 4)$
- (C) $3x(x + 4) + 4$
- (D) $(3x + 4)(x + 4)$
- (E) $(x + 4)(x + 4)$

Q3. Factor: $x^2 + 2x + 8x + 16$

- (A) $x(x + 2) + 8(x + 2)$
- (B) $x(x + 8) + 2(x + 8)$
- (C) $x(x + 2) + 8x$
- (D) $x^2 + 10x + 16$
- (E) $(x + 2)(x + 8)$

Q4. Factor the expression: $3x^2 + 12x + 2x + 8$

- (A) $3x(x + 4) + 2(x + 4)$
- (B) $3x(x + 2) + 4(x + 2)$
- (C) $(3x + 2)(x + 4)$
- (D) $3x(x + 4) + 2x + 4$
- (E) $(3x + 2)(x + 4)$

Q5. Verify factoring: $(x + 2)(x + 3)$ by multiplying

- (A) $x^2 + 5x + 6$
- (B) $x^2 + 2x + 6$
- (C) $x^2 + 6x + 5$
- (D) $x^2 + x + 6$
- (E) $x^2 + 4x + 6$

Q6. Factor the expression: $x^2 + x + 6x + 6$

- (A) $x(x + 1) + 6(x + 1)$
- (B) $x(x + 6) + 1(x + 6)$
- (C) $(x + 1)(x + 6)$
- (D) $(x + 6)(x - 1)$
- (E) $x^2 + 7x + 6$

Q7. Factor the expression: $2x^2 + 5x + x + 5/2$

- (A) $2x(x + 5/2) + 1(x + 5/2)$
- (B) $2x(x + 1/2) + 5(x + 1/2)$
- (C) $2x^2 + 6x + 5/2$
- (D) $(2x + 1)(x + 5/2)$
- (E) $(x + 5/2)(2x + 1)$

Q8. Factor: $2x^2 + 4x + x + 2$

- (A) $(x + 2)(2x + 1)$
- (B) $(2x + 1)(x + 2)$
- (C) $2x(x + 2) + x + 2$
- (D) $2x(x + 1) + 2(x + 1)$
- (E) $2x(x + 2) + 1(x + 2)$

Open Ended Questions (FRQ)

Q9. Factor the expression: $3x^2 + 12x + 4x + 16$. Show your steps and simplify.

Q10. Factor and verify by multiplying the following: $(2x + 3)(x + 2)$. Write final answer in factored form.

Chef's Tips

1

- Tip 1: Remind students that when factoring by grouping, it is essential to have a common factor in each group.
- Tip 2: Make sure students can apply the distributive property in reverse when verifying factoring by multiplication.
- Tip 3: Encourage students to use the FOIL method to multiply and verify their factorization.